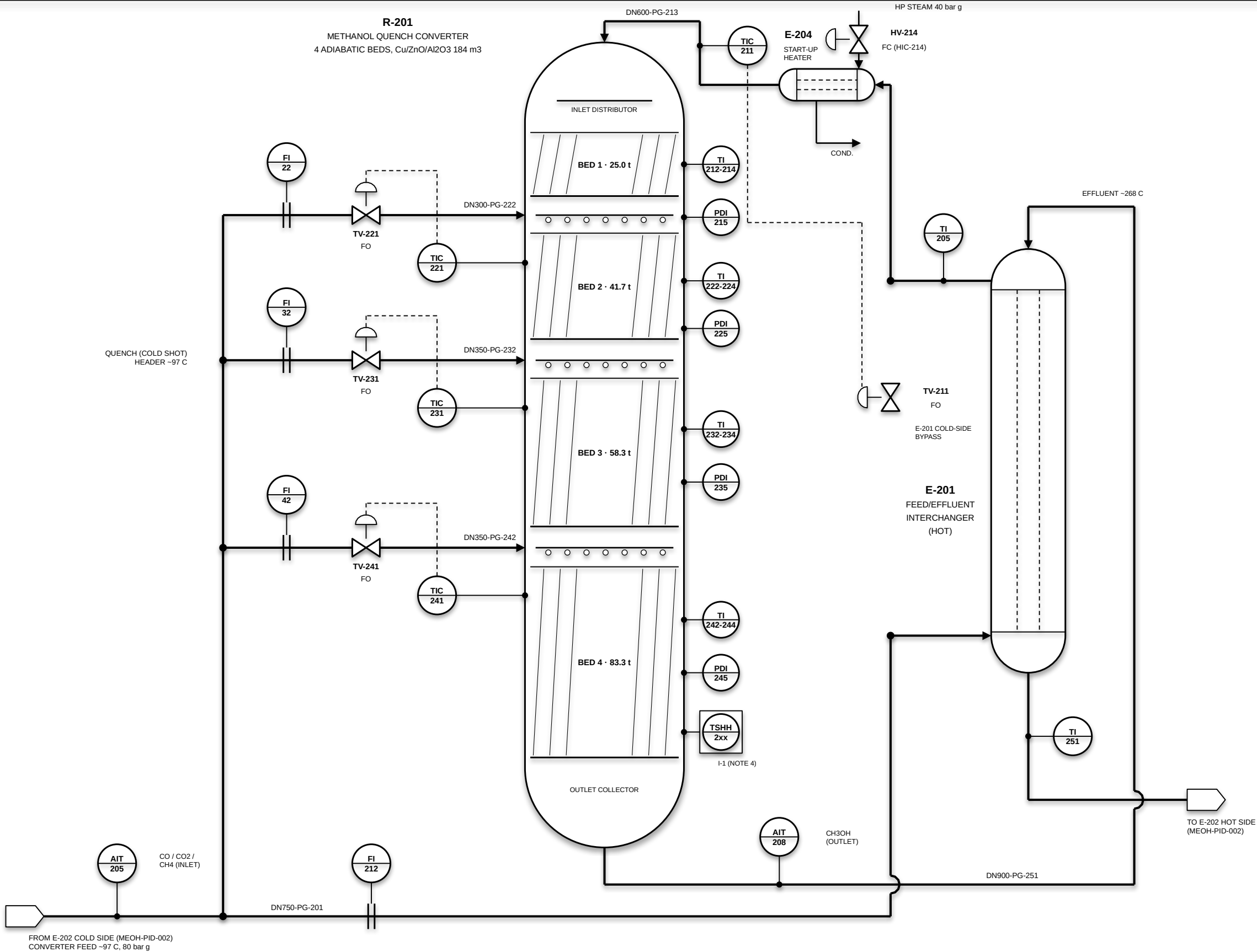
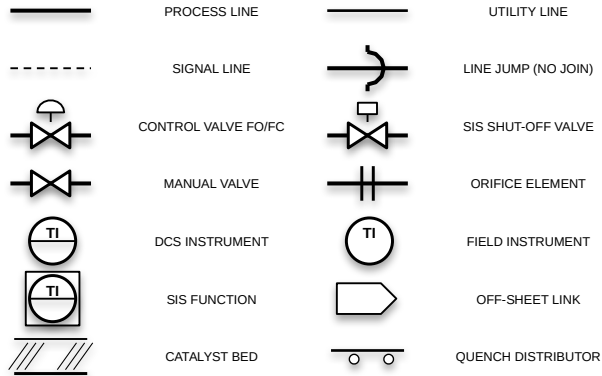


R-201
METHANOL QUENCH CONVERTER
4 ADIABATIC BEDS, Cu/ZnO/Al2O3 184 m3



LEGEND



CONTROL LOOP SCHEDULE

LOOP	MEASURES	MOVES	FAIL	SET / NOTE
FIC-201	CONVERTER FEED	C-201 SPEED (SC)	-	1928 kNm3/h
FIC-202	MAKE-UP SYNGAS	FV-202	FC	366 kNm3/h (RATE)
PIC-203	LOOP PRESS. V-201	PV-203 PURGE	FC	75.0 bar g
LIC-204	V-201 LEVEL	LV-204	FC	50 %
TIC-211	BED 1 INLET	TV-211 (E-201 BYP.)	FO	220 C
TIC-221	BED 2 INLET	TV-221 QUENCH 2	FO	225 C
TIC-231	BED 3 INLET	TV-231 QUENCH 3	FO	230 C
TIC-241	BED 4 INLET	TV-241 QUENCH 4	FO	235 C
HIC-214	-	E-204 STEAM	FC	START-UP ONLY

SIS INTERLOCKS (ACTION: MAKE-UP CUT)

I-	INITIATOR	SETTING	ACTION
1	TSHH-2xx ANY BED TC	> 300 C, 5 s	XV-202 + E-204 OFF
2	FSL-201 CIRCULATION	< 45 %, 5 s *	XV-202 + E-204 OFF
3	LSHH-204 V-201 LEVEL	> 90 %, 5 s	XV-202 + STOP C-201
4	PSHH-203 LOOP PRESS.	> 85 bar g, 5 s	XV-202 + E-204 OFF
5	HS-200 ESD PUSHBUTTON	MANUAL	XV-202 + C-201 + E-204

DESIGN POINT (OTS MODEL)

BED	IN C	OUT C	MeOH %	CAT. t
1	220	260	2.1	25.0
2	225	277	3.6	41.7
3	230	270	4.4	58.3
4	235	268	4.8	83.3

CONVERTER 80 bar g IN; V-201 75 bar g / 40 C; FEED 1928 kNm3/h;
MAKE-UP 366 kNm3/h; PURGE 105 kNm3/h; MeOH 112.5 t/h; Q(RXN) 85 MW.
* ARMED ONLY WITH XV-202 OPEN.

GENERAL NOTES:

- DESIGN BASIS (REFERENCE CARD): 900 kty CH3OH AT 8000 h/y = 112.5 t/h; QUENCH CONVERTER, FIXED BED, GAS PHASE, 220-280 C, 50-100 bar, Cu/ZnO/Al2O3, GHSV 8000-12000 1/h. OTS DESIGN: 80 bar g INLET, 184 m3 CATALYST (12/20/28/40 %, 1150 kg/m3), GHSV 10 500 1/h, FEED 1928 kNm3/h; BED TEMPERATURES AND FLOWS FROM THE OTS HEAT AND MASS BALANCE (KINETICS VANDEN BUSSCHE & FROMENT 1996, ASSUMED) - PLANT VALUES TBC.
- LINE SIZES PRELIMINARY (GAS 20 m/s, LIQUID 2 m/s) - TBC BY PIPING. SERVICE CODES: PG PROCESS GAS, SG SYNGAS, PU PURGE, CM CRUDE METHANOL, HS HP STEAM, SC CONDENSATE, CW COOLING WATER, FL FLARE.
- COLD SHOTS ENTER THROUGH LOZENGE DISTRIBUTORS ABOVE BEDS 2-4; TIC-221/231/241 MEASURE JUST BELOW EACH DISTRIBUTOR. QUENCH VALVES AND TV-211 FAIL OPEN (CONVERTER GOES COLDER ON AIR FAILURE).
- TSHH-2xx: ANY OF THE BED MULTIPOINT THERMOCOUPLES TE-212...244 ABOVE 300 C FOR 5 s (VOTING TBC) - I-1 CLOSSES XV-202 (MAKE-UP) AND FORCES HV-214 SHUT; C-201 KEEPS CIRCULATING TO COOL THE BEDS.
- E-204 START-UP HEATER (HP STEAM, MAX 22 MW) IN SERVICE ONLY TO HEAT THE CATALYST TO ~210 C BEFORE MAKE-UP GAS IS ADMITTED; HV-214 BLOCKED IN DURING NORMAL OPERATION.
- PDI-215...245 TAPS ABOVE AND BELOW EACH BED (NOT SHOWN). CONVERTER RELIEF IS PROVIDED BY PSV-201 ON V-201 (MEOH-PID-002); CATALYST DUMP NOZZLES, MANWAYS, VENTS AND DRAINS NOT SHOWN.
- TIC-211 SETS THE BED 1 INLET BY BYPASSING PART OF THE BED-1 FEED AROUND THE COLD SIDE OF E-201 (FEED/EFFLUENT POSITIVE FEEDBACK - TUNE SLOW, CHECK AFTER ANY RATE CHANGE).

REV	DESCRIPTION	DATE	
A	ISSUED FOR OTS DEVELOPMENT (REVIEW)	25-SEP-26	
CHULA CHEM. ENG. - METHANOL CONVERTER VER 1 (OTS)			
P&ID - METHANOL CONVERTER R-201, INTERCHANGER E-201, E-204			
DRAWN: APINAN SOOTTITANTAWAT	CHK'D: --	APP'D: --	DATE: 25-SEP-26
DWG NO. MEOH-PID-001	REV A	SCALE N.T.S.	SHEET 1 OF 2